Milestone 3 Recommender System Deliverables

Repository: https://github.com/Padmarao2024/milestone3
Successful CI/CD Run: https://github.com/Padmarao2024/milestone3/actions/runs/24795929105
Live Health Endpoint: https://recommender-api.gentlemushroom-c3c537f6.eastus.azurecontainerapps.io/health

1. Pipeline Structure & Config Separation

- Shared package layout cleanly separates config, quality, transform, train, serve, and eval in `pipeline/`.
- Environment-based configuration is centralized in `pipeline/config.py` instead of being duplicated across scripts.
- Thin operational entrypoints remain in `ingestor/`, `training/`, and `api/app/recommender.py`.
- Flow is explicit and reproducible: ingest -> transform -> train -> serialize -> serve -> eval.

Primary code links: `pipeline/config.py`, `pipeline/quality.py`, `pipeline/transform.py`, `pipeline/train.py`, `pipeline/serve.py`, `pipeline/eval.py`

2. Offline Evaluation Spec & Reproducibility

- Split: per-user chronological leave-last holdout to preserve temporal ordering.
- Leakage control: recommendation history uses only pre-holdout interactions per user.
- Metrics: HR@10 and NDCG@10, plus subpopulation analysis and benchmark support metrics.
- Reproducible entrypoints and artifacts: `training/evaluate.py`, `report/offline_metrics.csv`, `report/offline_subpopulations.csv`, `report/benchmark.csv`.

Model	HR@10	NDCG@10	Users
als	0.3333	0.1667	3
item_item	0.3333	0.2103	3
popularity	0.3333	0.1667	3

Subpopulation highlight: item-item reaches HR@10 = 1.0 and NDCG@10 = 0.6309 on warm users; all models are weak on the hot bucket in this small sample.

3. Online KPI from Kafka Logs

- Source topic: alpha.reco_responses.
- Proxy success definition: status_code == 200, num_results >= 1, and latency_ms <= 2000.
- Supporting KPIs: healthy_response_rate, personalized_rate, error_rate, latency_ms_p95.
- Implementation and outputs: `pipeline/eval.py`, `training/online_eval.py`, `report/online_kpi_summary.csv`, `report/online_kpi_by_model.csv`.

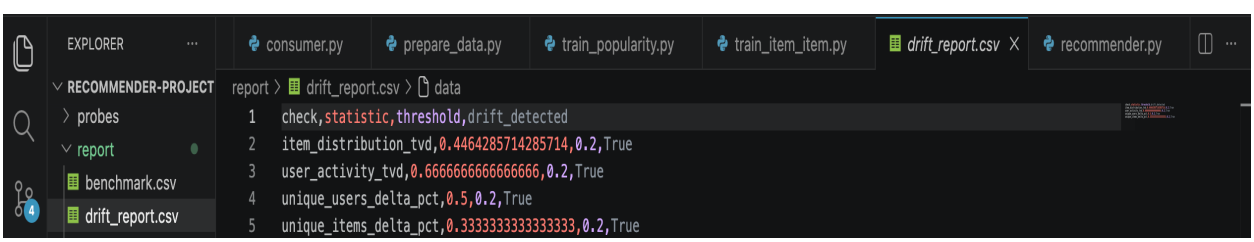
Responses	Proxy Success	Healthy	Personalized	P95 Latency	Error
6	83.33%	83.33%	50.00%	15381 ms	16.67%

Model	Responses	Proxy Success Rate	Personalized Rate
error	1	0.00	0.00
item_item	3	1.00	1.00
popularity	2	1.00	0.00

4. Data Quality, Drift, and Backpressure

- Schemas implemented with Pandera in `pipeline/quality.py` and enforced in `ingestor/validator.py`.
- Validated payload families: watch, rate, and recommendation response events.
- Backpressure handling is implemented through BackpressureBuffer in `pipeline/quality.py` and used by `ingestor/consumer.py`.
- Drift output is captured in `report/drift_report.csv` as a compact evidence table.

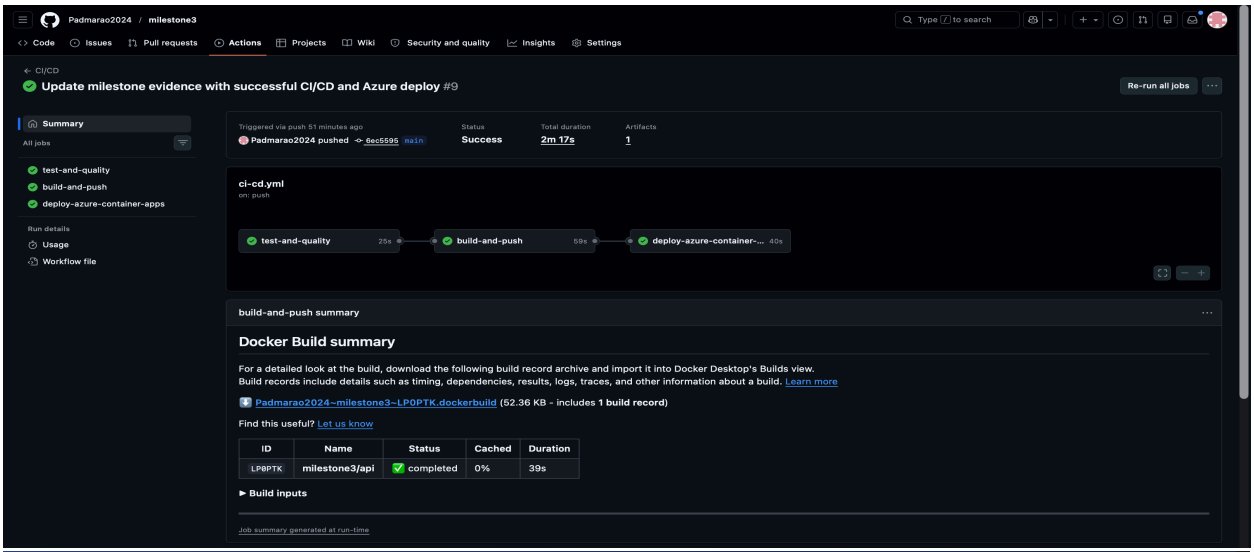
Drift Check	Statistic	Threshold	Drift
item_distribution_tvd	0.4464	0.20	True
user_activity_tvd	0.6667	0.20	True
unique_users_delta_pct	0.5000	0.20	True
unique_items_delta_pct	0.3333	0.20	True

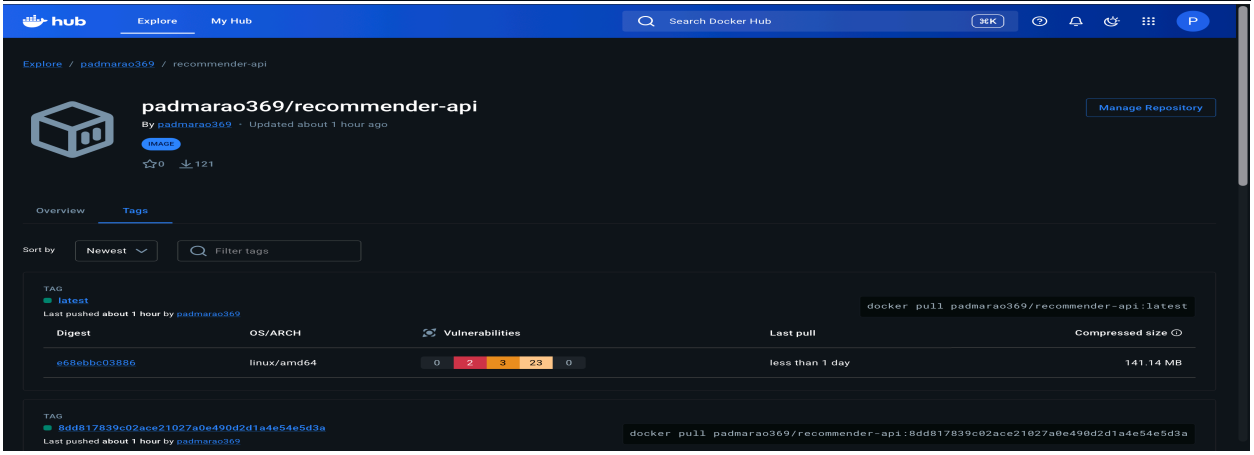


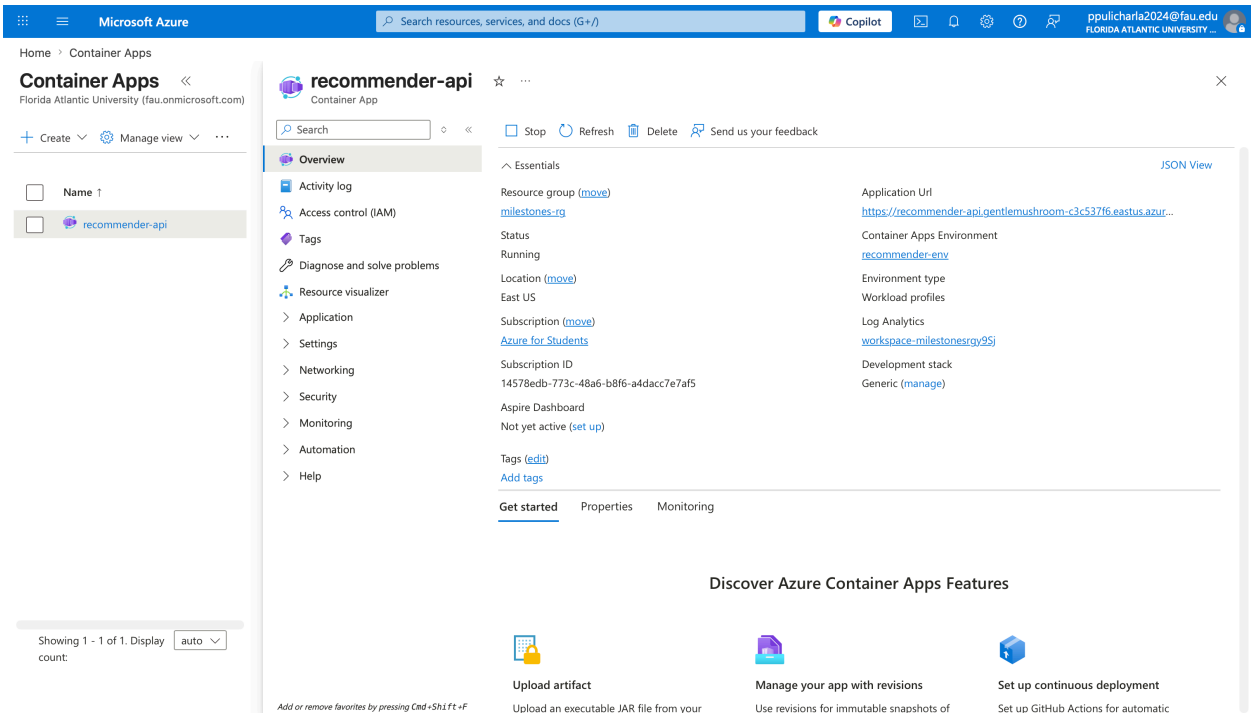
5. CI/CD, Deployment, and Test Report

- Workflow files: `.github/workflows/ci-cd.yml` and `.github/workflows/probe.yml`.
- Latest successful run contains green jobs for test-and-quality, build-and-push, and deploy-azure-container-apps.
- Secrets strategy: DOCKERHUB_USERNAME, DOCKERHUB_TOKEN, AZURE_CREDENTIALS, AZURE_CONTAINER_APP_NAME,AZURE_RESOURCE_GROUP.
- Deployment target is Azure Container Apps; current app health is HTTP 200 at the linked health endpoint.
- Test summary from `report/test_report.txt`: 10 passed, 74.68% coverage on scoped glue modules.

Requirement	Status	Evidence
CI checks	PASS	Lint + pytest coverage gate green
Docker build/push	PASS	Docker Hub image built from main
Azure deploy	PASS	Container App running and healthy
Repo hygiene	PASS	README, workflows, report artifacts present








```
{"status":"ok"}
```

Code and Evidence Links

- Offline code: `pipeline/eval.py`, `training/evaluate.py`
- Online KPI code: `training/online_eval.py`, `report/online_kpi_summary.csv`
- Quality gates: `pipeline/quality.py`, `ingestor/validator.py`, `ingestor/consumer.py`
- CI/CD and evidence: `.github/workflows/ci-cd.yml`, `report/milestone3_evidence.md`, `report/test_report.txt`
- Live deployment proof: https://recommender-api.gentlemushroom-c3c537f6.eastus.azurecontainerapps.io/health